

Progressive Visual Analytics for EEG Artifact Detection

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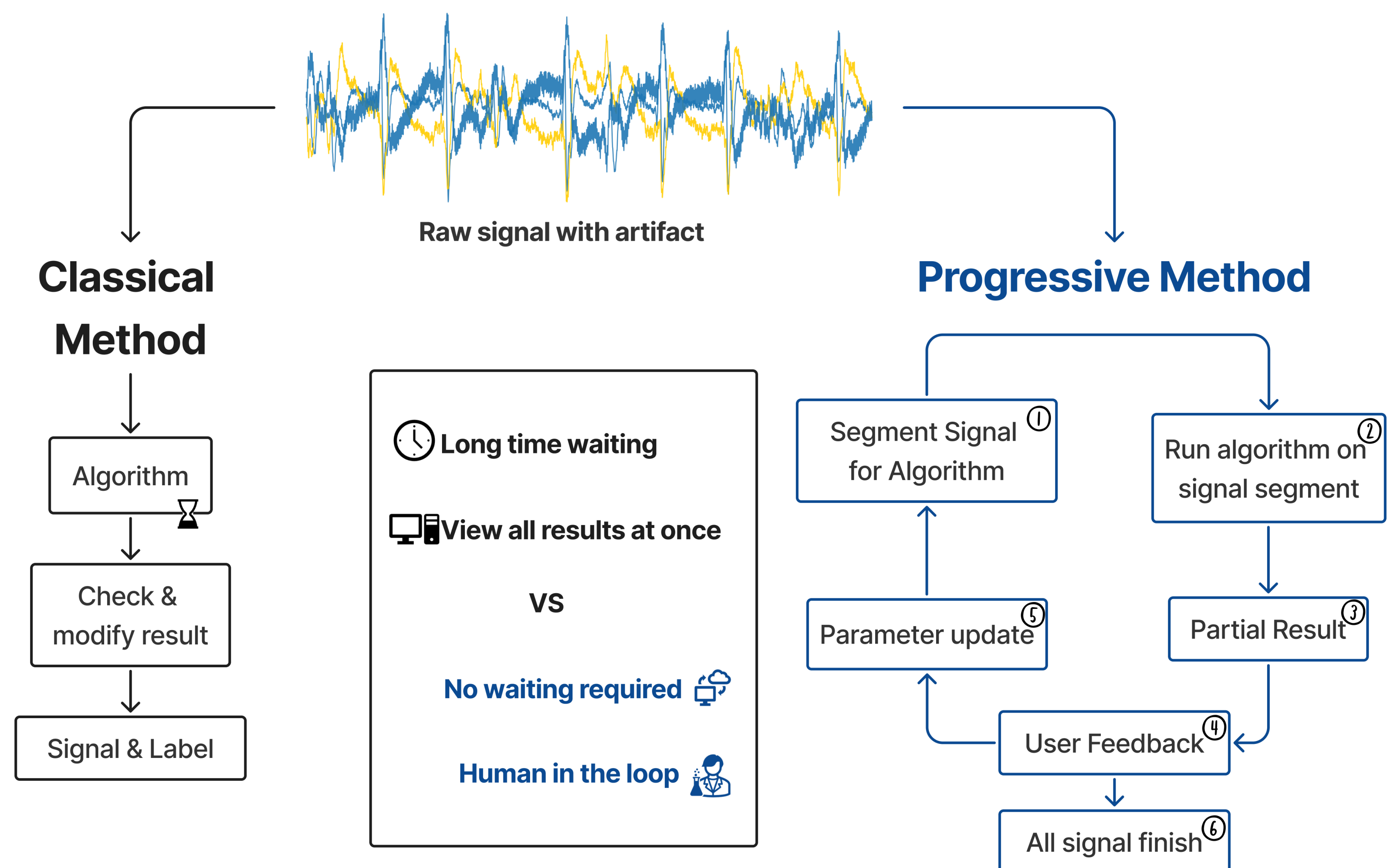


Problem Statement

Electroencephalography (EEG) artifact is the noise during signal collection, limiting the performance of the EEG application [3]. The signal is long, and the artifacts are located randomly in the long term, making them difficult to identify and annotate accurately. However, the EEG signal is inherently unstable, exhibiting significant variability over time and across individuals. These non-stationary characteristics limit the effectiveness of static models and fixed preprocessing pipelines.

To address these challenges, we adopt a Progressive Visual Analytics (PVA) approach, which enables users to engage in early-stage analysis, iterative refinement, and interactive decision-making. Rather than waiting for full computations to complete, users can interact with intermediate outputs, guiding the analysis as it unfolds. This is particularly well-suited for window-based artifact detection algorithms, which operate on segmented portions of the EEG signal. These algorithms naturally produce results incrementally, making them compatible with progressive interaction.

Our system leverages PVA to enhance an ongoing EEG artifact detection with ground-truth information provided by the human user on the intermediate outputs. This way, the algorithms can take the human-provided ground truth into account and use it to refine their detection while they are still running.

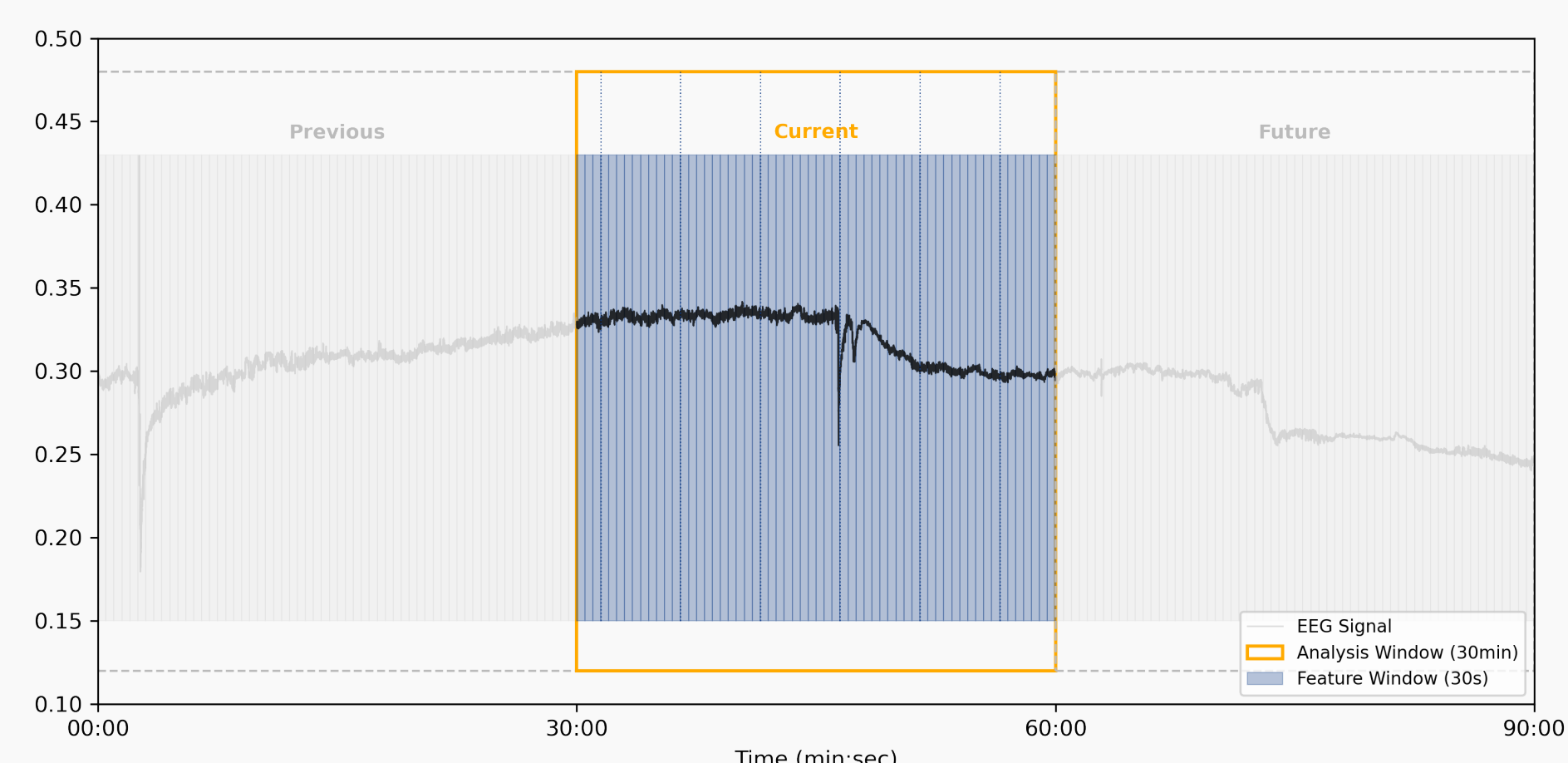


Progressive Visual Analytics Approach

Data Split for Progressive Strategy

Feature Window: The smallest unit of signal segment on which the algorithm computes features to assess artifact presence.

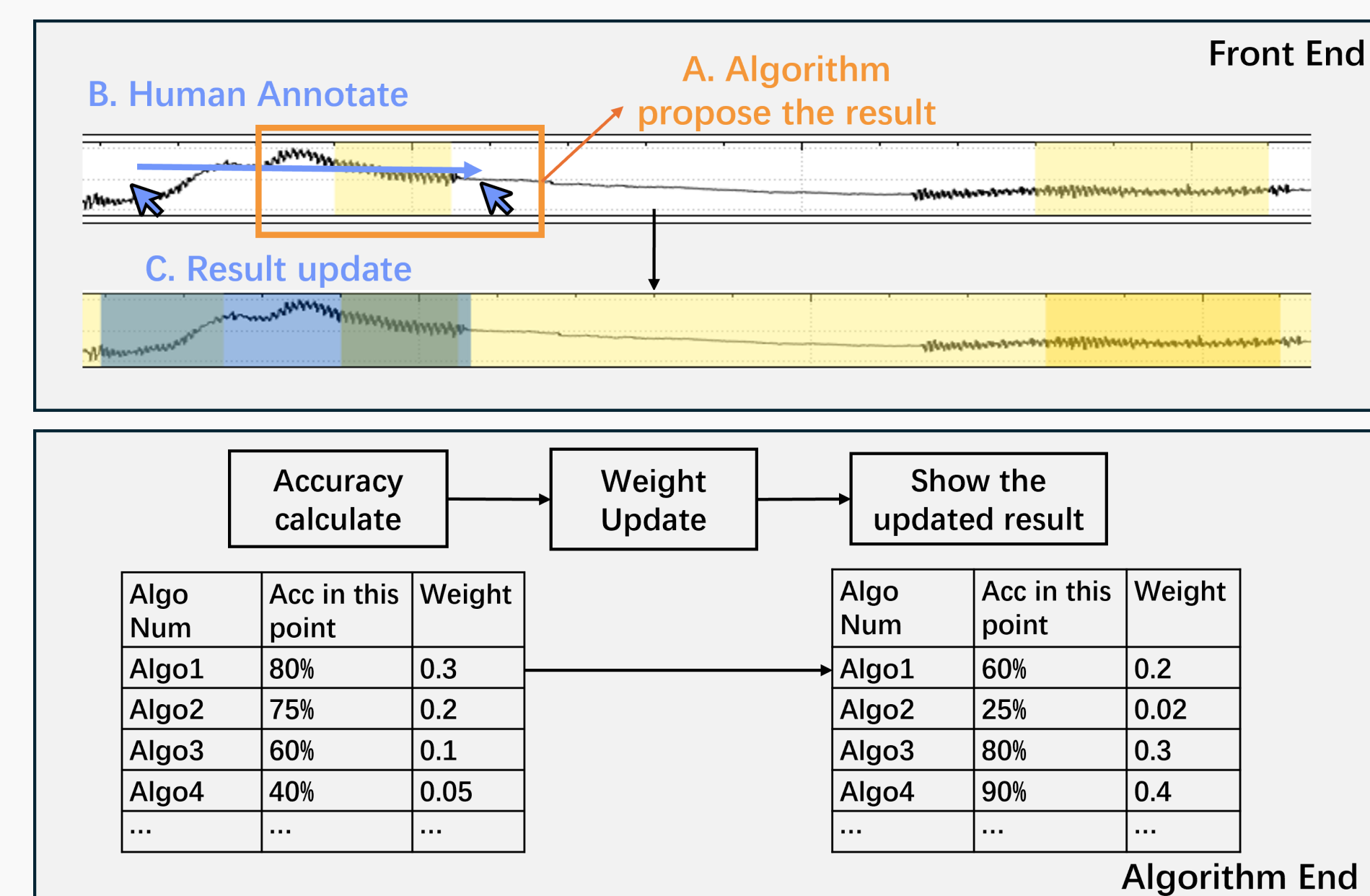
Analysis Window: A larger segment that contains several feature windows and serves as the decision unit for one round of classification.



*Feature window size is only for illustration purposes and does not reflect the actual window length used in the algorithm.

Multiple algorithms run in the current analysis window in parallel. After the algorithms finish classification in the current analysis window, the results from the multi-algorithms are fused and presented to users. Compared with the classical method, users do not need to wait for the algorithm to finish all the signal classification; they only need to focus on the current analysis window. Every time the user finishes interacting with the current window, the multi-algorithms improve themselves with the human action.

Human-Multiple Algorithm Interaction



Algorithm Fusion: The final decision for each feature window is determined by a weighted majority vote, where the weight of each algorithm is calculated as:

$$w_i = \frac{B\text{Acc}_i}{\sum_j B\text{Acc}_j}$$

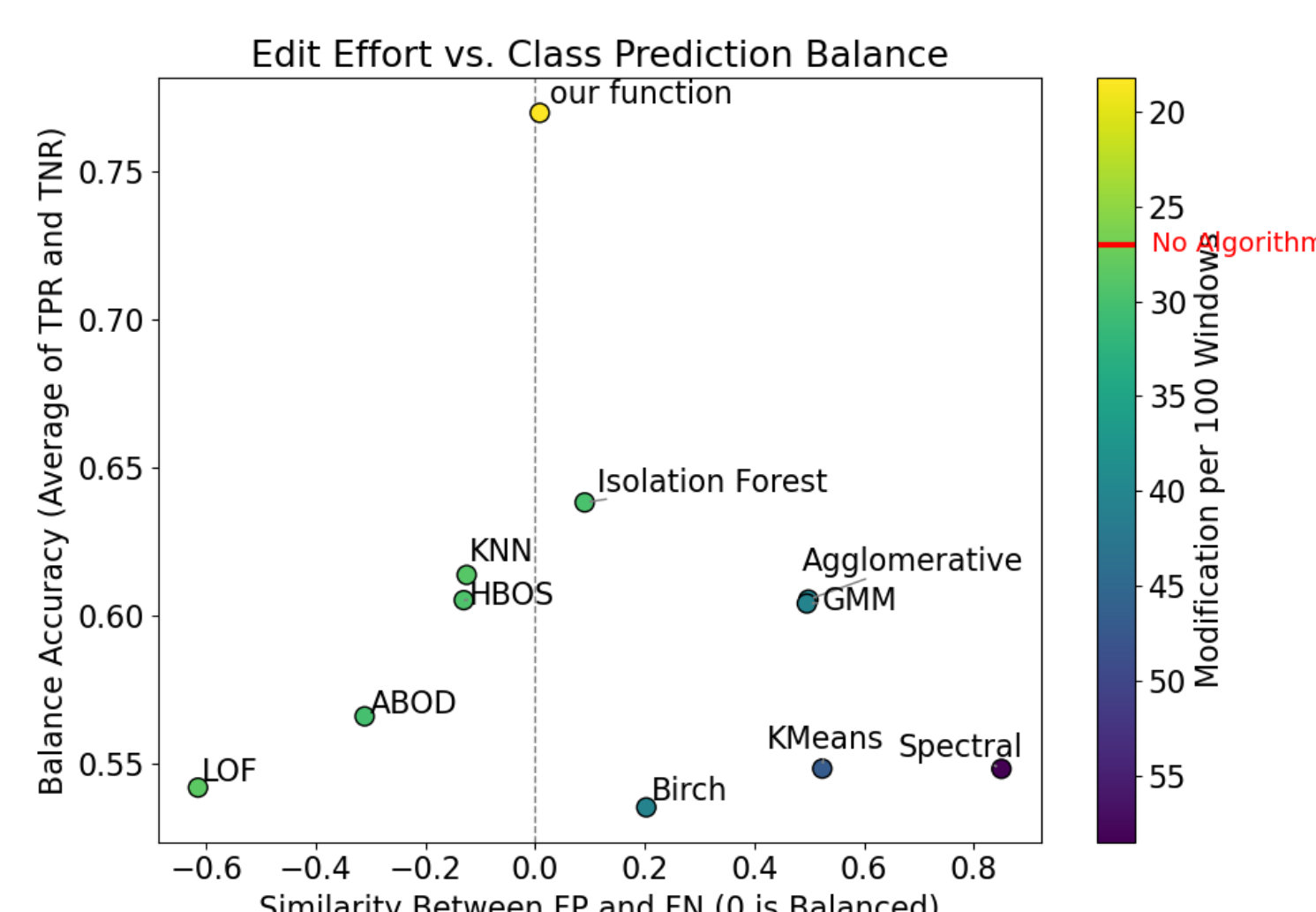
with $B\text{Acc}_i$ representing the balanced accuracy of algorithm i computed from recent human annotations.

Human in the Loop: Users can annotate artifact regions at any time. These annotations are used to evaluate each algorithm's balanced accuracy in the backend and influence the fusion weights in real time.

Evaluation

The progressive interaction not only reduces user waiting time and improves usability, but also enhances the final classification accuracy through the fusion of multiple algorithm results by using human annotation as the reference. We used these three different metrics to validate the approach's performance:

1. Similarity = $\frac{FP-FN}{FP+FN}$ — measures prediction bias; 0 means FP and FN are balanced
2. Balanced Accuracy = $\frac{1}{2} \left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP} \right)$ — average performance across both classes
3. Modification per 100 Windows — how many errors the user needs to tune on average for 100 feature windows



Our function has the highest balance accuracy compared with each single unsupervised algorithm. And the FP and FN do not have a high similarity, which means our function does not have a bias towards one class. And our function has the smallest number of required edits, which is even smaller than the user can annotate every artifact window precisely without the algorithm.

References

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